

AI FITTRACK API

Proposed Solution

Functional design of AI FitTrack backend

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DATE	30 September 2026

Overview

A modular JSON REST service authenticates users, persists workouts in MongoDB, supports search, and calls Gemini for guidance.

Modules

- config/db.js connects MongoDB.
- app.js configures Express middleware and routes.
- Routes map auth, workout, and AI endpoints.
- Controllers validate and coordinate.
- Models define User and Workout.
- Middleware protects routes and formats errors.
- geminiService isolates provider calls.

Security

Passwords are hashed, tokens verified, and workout ownership checked. Keep credentials in local environment configuration.

Out of scope

Frontend, wearables, notifications, billing, clinical advice, and deployment are not included.